

NORTH SYDNEY GIRLS HIGH SCHOOL

EXTENSION 2 MATHEMATICS

HSC ASSESSMENT TASK

TERM 1 2007

Name: _____ **Class:** _____

Instructions:

- Time allowed: 60 minutes + 2 minutes reading time.
- This task is worth 20% of your HSC assessment
- Attempt all three questions
- Show all necessary working
- Marks may be deducted for incomplete or poorly arranged work
- Start each question on a NEW PAGE
- Write on one side of the page only

Question1 (15 marks)**Marks**

(a) Given the complex number $z = 3 - 3i$ write down in exact form

(i) $|z|$ 1

(ii) $\arg z$ 1

(b) Express in the form $x + iy$ where x and y are real

(i) $(7 + 3i)(\overline{4 - i})$ 1

(ii) $\left(\frac{2 - 5i}{4 - 3i}\right)$ 2

(c) Evaluate $\sqrt{9 + 40i}$ 2

(d) Find real numbers for a and b if $z = 3 + i$ satisfies $z + \frac{a}{z} = b$ 3

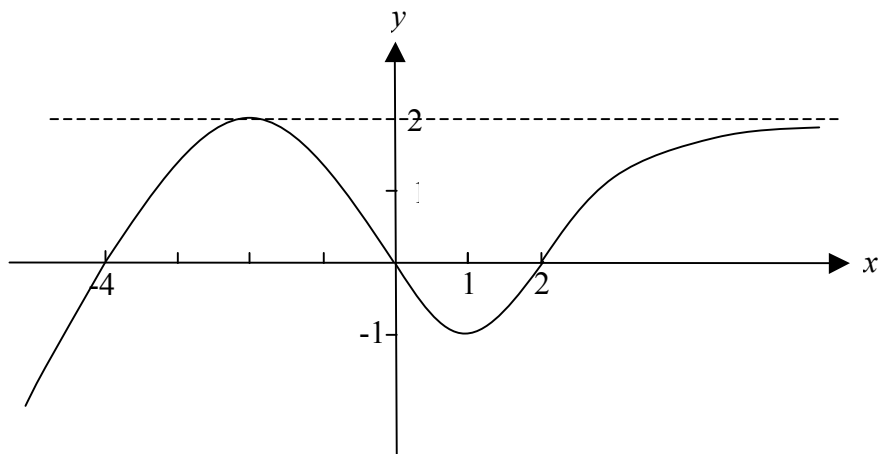
(e) Sketch the region on the Argand plane where 2

$$|z - 3| < 3 \quad \text{and} \quad 0 < \arg z < \frac{\pi}{4}$$

(f) Describe in both algebraic and geometric terms the set of points z in the Argand plane such that $z^2 + (\bar{z})^2 = 0$ 3

Question 2 (15 marks)**Marks**

- (a) The graph of $y = f(x)$ is shown below



Draw neat sketches of the following *on the answer sheet provided*, showing clearly any stationary points and asymptotes

- | | | |
|-------|---------------------------------|---|
| (i) | $y = \frac{1}{f(x)}$ | 2 |
| (ii) | $y = f\left(\frac{x}{2}\right)$ | 2 |
| (iii) | $y = [f(x)]^2$ | 2 |
| (iv) | $y = f(x) + f(x) $ | 2 |
| (v) | $y = 2^{f(x)}$ | 2 |

- (b) Consider the function $f(x) = x(x-3)^2$

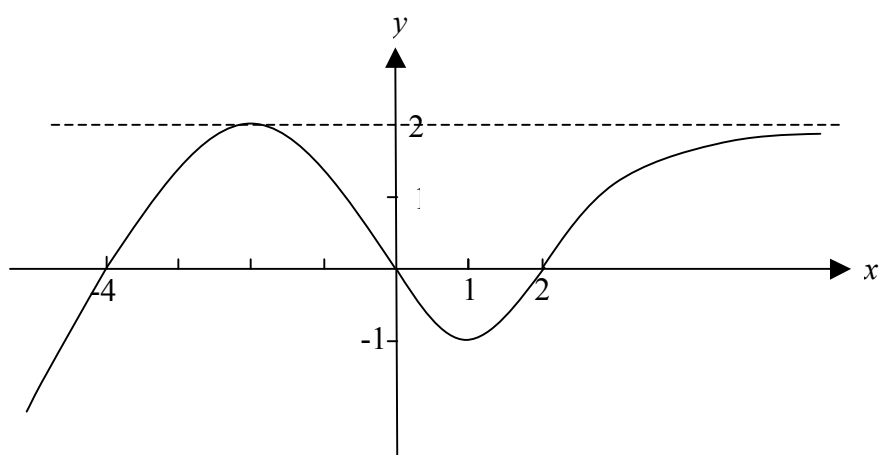
- | | | |
|-------|--|---|
| (i) | Sketch the graph of the curve showing the coordinates of any turning points and the intercepts with the coordinate axes. | 2 |
| (ii) | Hence draw a neat separate sketch of $y^2 = f(x)$ | 2 |
| (iii) | Describe the nature of the curve $y^2 = f(x)$ at $x = 0$. | 1 |

Marks

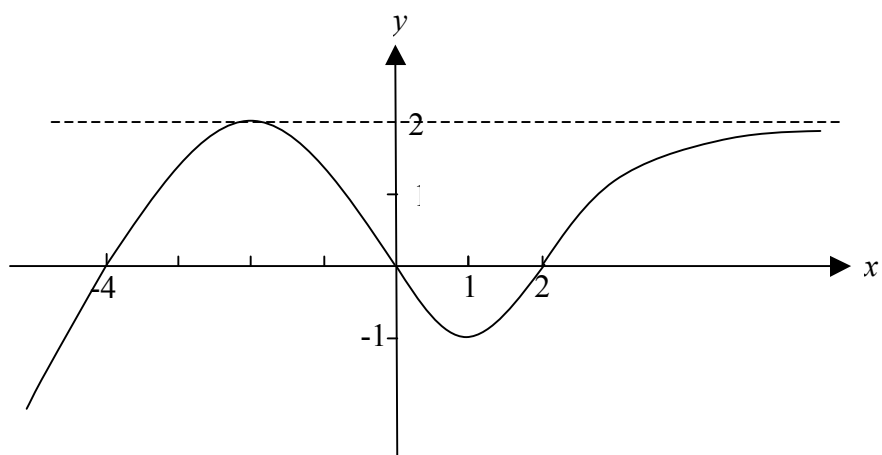
- END OF TEST**

Name: _____ Class: _____

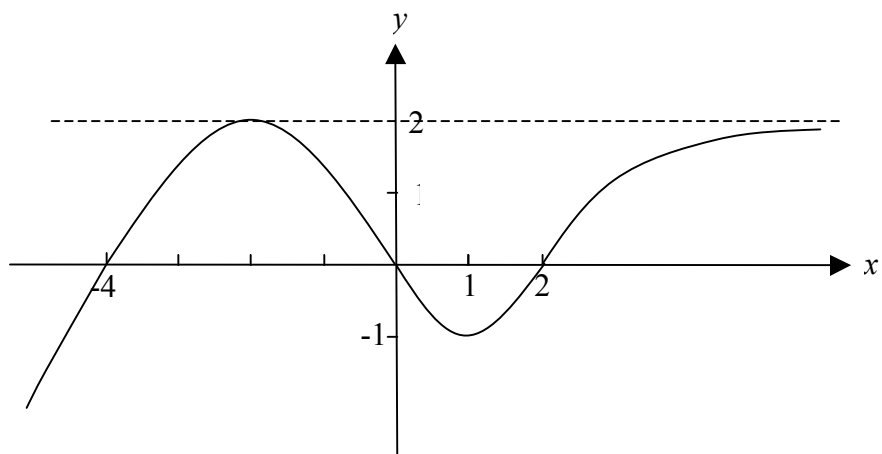
(i)



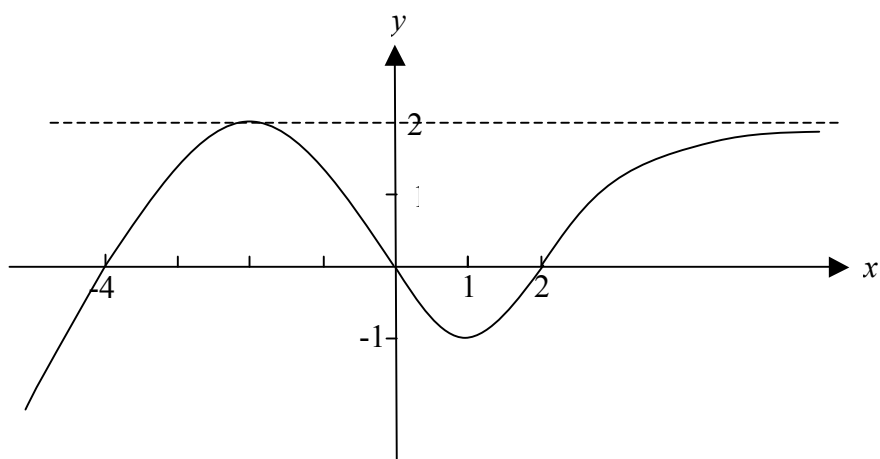
(ii)



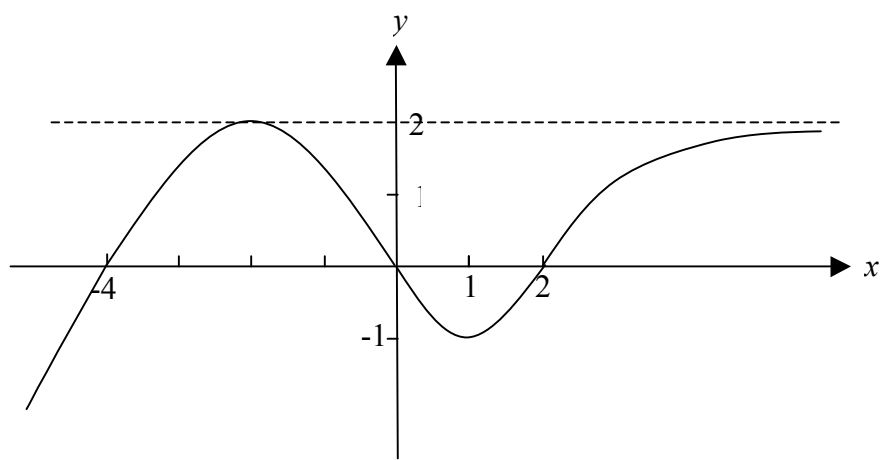
(iii)



(iv)



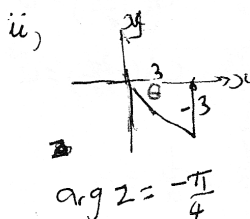
(v)



Question 1

(a) $z = 3 - 3i$

(i) $|z| = \sqrt{3^2 + (-3)^2}$
 $= \sqrt{18}$
 $= 3\sqrt{2}$



(b) $(z + 3i)(\overline{4 - i})$
 $= (7 + 3i)(4 + i)$
 $= 28 + 7i + 12i - 3$
 $= 25 + 19i$

(ii) $\left(\frac{2 - 5i}{4 - 3i}\right) \left(\frac{4 + 3i}{4 + 3i}\right)$
 $= \frac{8 + 6i - 20i + 15}{16 + 9}$
 $= \frac{23 - 14i}{25}$
 $= \frac{23}{25} - \frac{14}{25}i$

(c) let $z = x + iy = \sqrt{9 + 40i}$
 $(x + iy)^2 = 9 + 40i$
 $x^2 - y^2 = 9 \quad 2xy = 40$
 $x^2 - y^2 = 9 \quad xy = 20$
 $x = 5, y = 4$
 $\therefore 5 + 4i$

(d) $z + \frac{a}{z} = b$

$(3 + i) + \frac{a}{3 + i} = b$

$(3 + i)^2 + a = b(3 + i)$

$9 + 6i - 1 + a = 3b + ib$

$(8 + a) + 6i = 3b + ib$

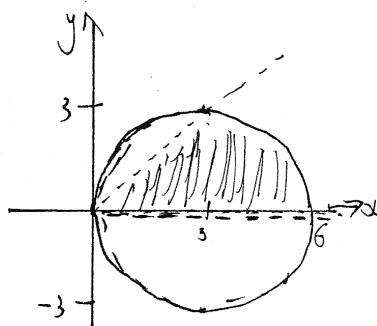
$\therefore b = 6$

$8 + a = 3b$

$8 + a = 18$

$a = 10$

(e) $|z - 3| < 3$ and $0 < \arg z < \frac{\pi}{4}$



(f) $z^2 + (\overline{z})^2 = 0$

let $z = x + iy$

$(x + iy)^2 + (x - iy)^2 = 0$

$x^2 + 2xyi + y^2 + x^2 - 2xyi + y^2 = 0$

$2x^2 - 2y^2 = 0$

$x^2 - y^2 = 0$

$(x - y)(x + y) = 0$

Pair of lines $y = \pm x$

Worksheet 2(a),
Question 2(b)

(i) $y = x(x - 3)^2$

$y' = (x - 3)^2 + x \cdot 2(x - 3)$

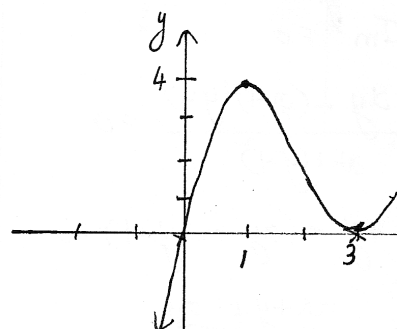
$= (x - 3)[x - 3 + 2x]$

$= (x - 3)(3x - 3)$

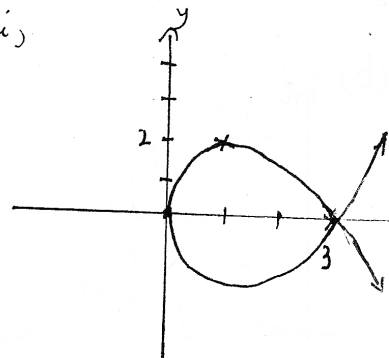
$= 3(x - 1)(x - 3)$

$x = 1, y = 4$

$x = 3, y = 0$



(ii)



Vertical tangent.

Question 3

$$(a) \operatorname{Im} \left(\frac{z-1}{z+i} \right) = 0 \quad (z \neq i)$$

$$\text{Let } z = x+iy$$

$$\left[\frac{(x-1)+iy}{x+i(y+1)} \right] \times \left[\frac{x-i(y+1)}{x-i(y+1)} \right]$$

$$= \frac{x(x-1)+iyx-i(x-1)(y+1)+y(y+1)}{x^2+(y+1)^2}$$

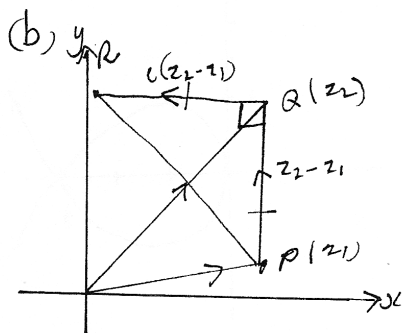
$$\operatorname{Im} \uparrow = 0$$

$$\frac{xy - (x-1)(y+1)}{x^2+(y+1)^2} = 0$$

$$xy - (xy + x - y - 1) = 0$$

$$-x + y + 1 = 0$$

$$\boxed{x - y + 1 = 0} \text{ is line.}$$



$i(z_2 - z_1)$ is vector $\vec{z_2 z_1}$
rotated 90° clockwise

$$z_1 + i(z_2 - z_1) \text{ is}$$

$(z_2 - z_1)$ 90° clockwise

added to vector oz_1

$\therefore \triangle OPQ$ right angled

$$\text{Q1, } z^5 = i$$

$$\text{Let } z = (\cos \theta + i \sin \theta)$$

$$\therefore (\cos \theta + i \sin \theta)^5 = i$$

$$\cos 5\theta + i \sin 5\theta = 0 + i$$

$$\cos 5\theta = 0$$

$$\sin 5\theta = 1$$

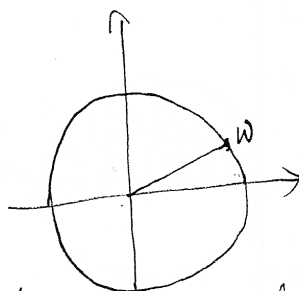
$$\therefore 5\theta = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{10}$$

$$\therefore \omega = \cos \frac{\pi}{10} + i \sin \frac{\pi}{10}$$

Smallest argument.

Q2



roots evenly spaced around unit circle

$$\text{ie arguments } \frac{\pi}{10} + \frac{2\pi}{5} = \frac{5\pi}{10} = \frac{\pi}{2}$$

$$\frac{5\pi}{10} + \frac{4\pi}{10} = \frac{9\pi}{10}$$

$$\frac{9\pi}{10} + \frac{4\pi}{10} = \frac{13\pi}{10}$$

$$\frac{13\pi}{10} + \frac{4\pi}{10} = \frac{17\pi}{10}$$

$$\text{ie } z_1 = \cos \frac{\pi}{10} = \omega$$

$$z_2 = \cos \frac{5\pi}{10} = \omega^5$$

$$z_3 = \cos \frac{9\pi}{10} = \omega^9$$

$$z_4 = \cos \frac{13\pi}{10} = \cos \left(\frac{-7\pi}{10} \right) = \omega^{-7}$$

$$z_5 = \cos \frac{17\pi}{10} = \cos \left(\frac{-3\pi}{10} \right) = \omega^{-3}$$

$$\text{Q3a) Sum of roots} = \frac{b}{a} = 0$$

$$\therefore \omega + \omega^5 + \omega^9 + \omega^{-7} + \omega^{-3} = 0$$

$$\omega^7 + \omega^3 + \omega + \omega^5 + \omega^9 = 0$$

(d)

$$(2 + 2i \tan \theta)^n + (2 - 2i \tan \theta)^n$$

$$= (2 + 2i \frac{\sin \theta}{\cos \theta})^n + (2 - 2i \frac{\sin \theta}{\cos \theta})^n$$

$$= 2^n \left[\frac{\cos \theta + i \sin \theta}{\cos \theta} \right]^n + 2^n \left[\frac{\cos \theta - i \sin \theta}{\cos \theta} \right]^n$$

$$= 2^n \left[\frac{\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta}{\cos^n \theta} \right]$$

$$= 2^n \cdot \frac{2 \cos n\theta}{\cos^n \theta}$$

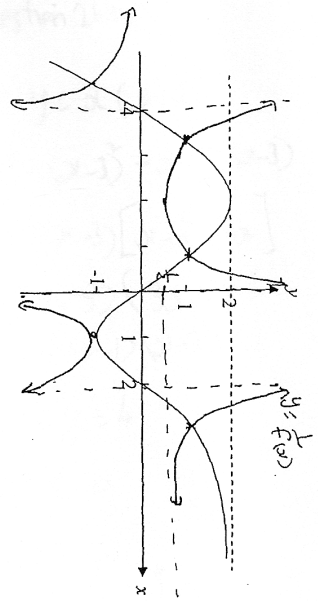
$$= \frac{2^{n+1} \cos n\theta}{\cos^n \theta}$$

$$= \frac{2^{n+1} \cos n\theta}{\cos^n \theta}$$

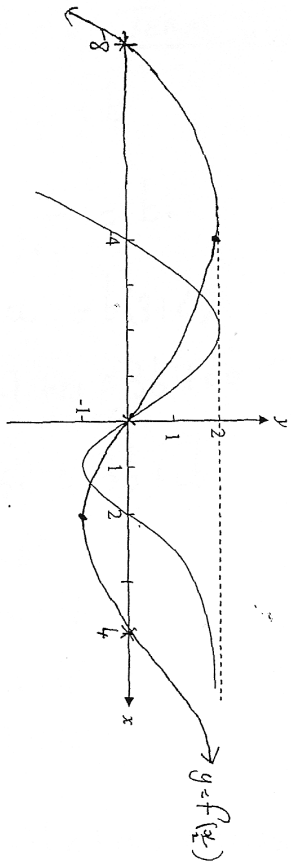
Name: _____
(i)

Q2 (a) Solutions.

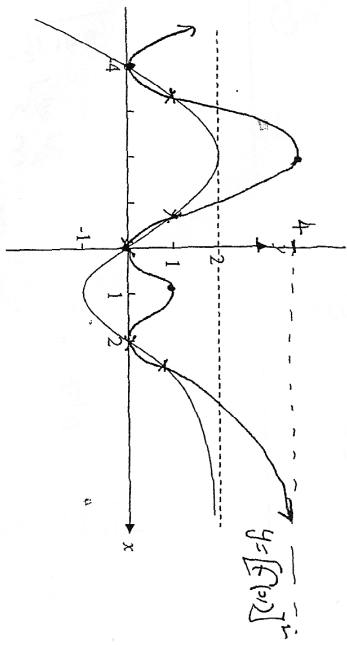
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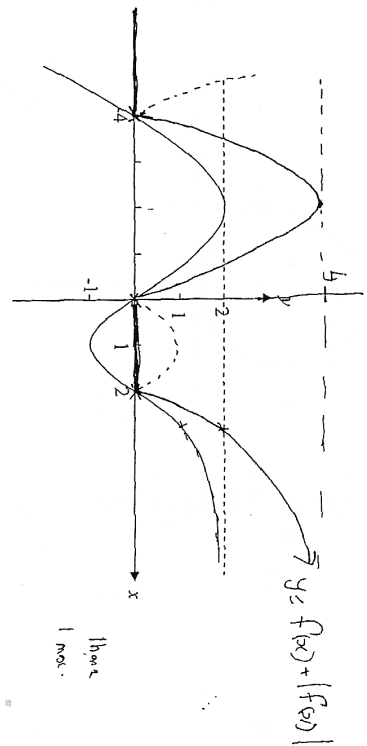
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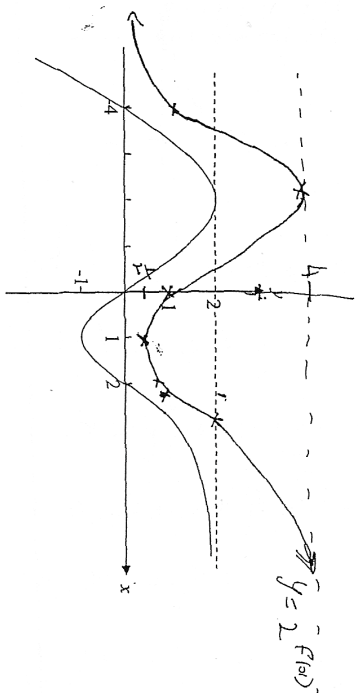
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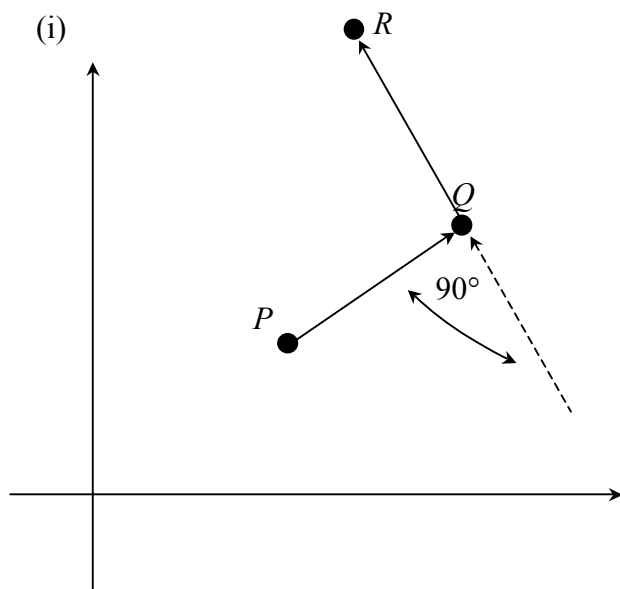
(v)



Q3

(b)

(i)



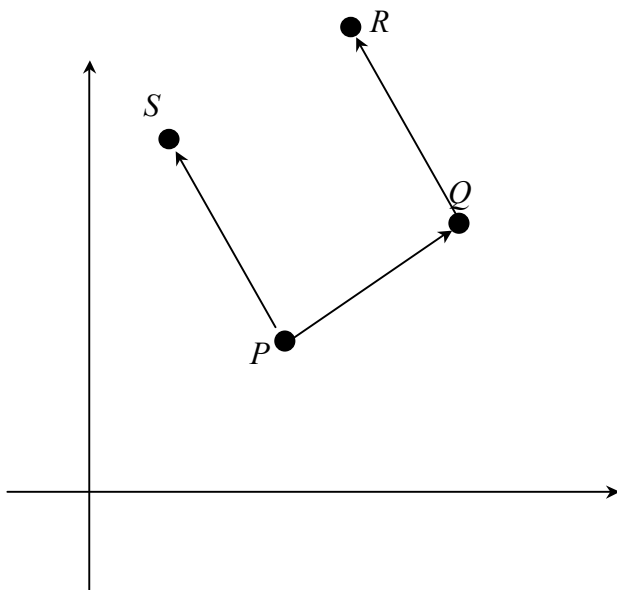
$$\overrightarrow{PQ} = z_2 - z_1$$

$$\overrightarrow{RQ} = z_2 + i(z_2 - z_1) - z_2 = i(z_2 - z_1)$$

$\therefore \overrightarrow{RQ}$ is an anti-clockwise rotation of $\overrightarrow{PQ}$ by 90°

So there is a right angle at Q making ΔPQR right angled and isosceles

(ii)



Let S be the point that represents the number z_3 , so that $PQRS$ is a rectangle (square)

$\therefore \overrightarrow{SP} = \overrightarrow{RQ}$ (opposite sides are equal and parallel)

$$\therefore z_3 - z_1 = i(z_2 - z_1)$$

$$\therefore z_3 = z_1 + i(z_2 - z_1)$$